Locality emerges from hidden zeros in $\mathrm{tr}(\phi^3)$: a by-hand proof at four, five, and six points

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Abstract

We prove that the cyclic hidden one-zeros of color-ordered $\mathrm{tr}(\phi^3)$ tree amplitudes force every non-local coefficient of the general rational ansatz to vanish, at $n = 4, 5, 6$ external legs. The proof is entirely by hand: it uses only the kinematic mesh identity, elementary geometric series Laurent expansions, and the fact that rational functions in independent variables vanish iff each coefficient on a basis of monomials vanishes. The geometric heart of the argument is the *enclosed-vertex principle*: each cyclic zero is tailor-built to kill non-local terms associated with chords crossing a designated "special" short diagonal. At $n = 6$ we give a complete orbit-by-orbit enumeration showing that 129 of the 151 non-local coefficients are killed by a "Layer 1" argument ($x_r \to \infty$ limit), and the remaining 22 are killed by "Layer 2" companion-isolation; the 22 fall into exactly 5 cyclic-symmetry orbits, and we verify one representative of each.

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1 Setup

1.1 Chords, the ansatz, and locality

Label the external legs cyclically by $1, \ldots, n$. The planar Mandelstam variables are

$$X_{ij} \;=\; (p_i + p_{i+1} + \cdots + p_{j-1})^2, \qquad 1 \le i < j \le n,\ j - i \ge 2,\ (i,j) \ne (1,n).$$

Geometrically, X_{ij} is the chord of the convex n-gon from vertex i to vertex j. There are $n(n-3)/2$ chords: $2, 5, 9$ for $n = 4, 5, 6$.

A *triangulation* of the n-gon is a maximal set of pairwise non-crossing chords; it has $n-3$ chords. The color-ordered tree amplitude of $\mathrm{tr}(\phi^3)$ is

$$A_n \;=\; \sum_{T \text{ triangulation}} \prod_{(i,j)\in T} \frac{1}{X_{ij}}.$$

The number of triangulations is the Catalan number C_{n-2}, which takes the values $2, 5, 14$ for $n = 4, 5, 6$.

We consider the most general rational function B with the correct mass dimension $-2(n-3)$ and only planar simple poles:

$$B \;=\; \sum_{\text{multisets } (a_1,\ldots,a_{n-3})} \frac{a_{a_1\ldots a_{n-3}}}{X_{a_1}\cdots X_{a_{n-3}}},$$

the sum running over multisets (size $n-3$, with repetition) of chord indices. With $d = n(n-3)/2$ chords, there are $\binom{d+n-4}{n-3}$ such multisets, giving $2, 15, 165$ coefficients for $n = 4, 5, 6$ respectively.

Definition 1. A multiset of size $n-3$ is *local* if its chord indices form a triangulation. Otherwise it is *non-local*: either it contains a repeated chord (higher-order pole) or two of its chords cross.

Locality is the statement that all non-local coefficients of B vanish.

1.2 The kinematic mesh identity

The non-planar two-particle invariants $c_{a,b} = 2p_a \cdot p_b$ (for $a \ne b$ non-adjacent) satisfy the *mesh identity*

$$c_{a,b} \;=\; X_{ab+1} + X_{a+1b} - X_{ab} - X_{a+1b+1}, \tag{1}$$

with the convention that adjacent chords like X_{aa+1} vanish (massless particles). Each equation $c_{a,b} = 0$ thus imposes a linear relation on the planar chords.

1.3 The hidden one-zeros

For each leg $r \in \{1, \ldots, n\}$, the *hidden one-zero* $\mathcal{Z}_r$ is the subvariety of kinematic space defined by the vanishing of all non-planar c-invariants involving leg r:

$$\mathcal{Z}_r : \qquad c_{r,r+2} = c_{r,r+3} = \cdots = c_{r,r+n-2} = 0 \pmod{n}.$$

This imposes $n-3$ linear constraints on the planar chords.

The starting fact (proved elsewhere and taken as given here) is that the tree amplitude A_n vanishes identically on each $\mathcal{Z}_r$. Our problem is the converse for the general ansatz B: assuming $B|_{\mathcal{Z}_r} = 0$ for all $r = 1, \ldots, n$, show that all non-local a-coefficients are forced to vanish.

1.4 The geometric heart: enclosed vertices

Every chord X_{ij} divides the polygon's boundary into two arcs. The *enclosed arc* is the shorter of the two, and the *enclosed vertex set* is

$$E(X_{ij}) \;=\; \{\text{vertices strictly between } i \text{ and } j \text{ on the shorter arc}\}.$$

We call $|E(X_{ij})|$ the *class* of the chord: class 1 chords ("short diagonals") enclose 1 vertex, class 2 ("long diagonals") enclose 2, and so on.

Lemma 1 (Crossing = enclosed-vertex sweep)**.** *Two chords X_{ij} and X_{rs} of a convex polygon cross in the interior iff exactly one endpoint of X_{rs} lies in $E(X_{ij})$, the other in the outer arc.*

Proof. Two chords of a convex polygon cross iff their four endpoints alternate around the boundary. Fix X_{ij} and consider the two arcs it cuts off. The endpoints of X_{rs} either both lie in the same arc (no alternation, no crossing) or one lies in each (alternation, crossing). □

Consequence. Every non-local pair of chords (crossing pair) has the form "one chord with enclosed vertex v, another chord starting at v." The hidden zeros, as we will see, are tailor-built to kill these non-local pairings.

2 The main theorem, executive summary

Theorem 1 (Locality at four, five, and six points)**.** *The cyclic hidden one-zeros $\mathcal{Z}_1, \ldots, \mathcal{Z}_n$ force every non-local coefficient of the general rational ansatz B to zero, at $n = 4, 5, 6$. The surviving C_{n-2} coefficients are exactly the tree triangulations of the n-gon.*

Our proof at $n = 4$ is trivial (no non-local terms exist); at $n = 5$ it is a direct computation over each of 5 zones; and at $n = 6$ it requires an orbit enumeration of the 151 non-local multisets:

Proposition 1 ($n = 6$ coverage summary)**.** *The* 151 *non-local multisets at $n = 6$ split into:*

- 129 *covered by "Layer 1" (the $x_r \to \infty$ limit at some $\mathcal{Z}_r$),*
- 22 *covered by "Layer 2" (companion-isolation at order $1/x_r^2$),*
- 0 *requiring higher layers.*

The 22 *Layer-1 failures form exactly* 5 *cyclic orbits:*

Orbit	*Size*	*Representative*	*Killing zone*
$(3,0)$ *squared distance-3 pairs*	*6*	$x_1^2\, x_4$	$\mathcal{Z}_6$
$(3,0)$ *non-consec triples, orbit A*	*6*	$\{x_1, x_2, x_4\}$	$\mathcal{Z}_6$
$(3,0)$ *non-consec triples, orbit B*	*6*	$\{x_1, x_2, x_5\}$	$\mathcal{Z}_4$
$(2,1)$ *"y-fork"*	*3*	$\{x_3, x_6, y_1\}$	$\mathcal{Z}_2$
$(0,3)$ *all-y*	*1*	$\{y_1, y_2, y_3\}$	$\mathcal{Z}_1$

The notation (p,q) refers to the chord-class composition: p short, q long.

The rest of the paper proves Theorem 1 by going through each of $n = 4, 5, 6$ carefully, and in particular verifying Proposition 1 with a complete orbit-by-orbit analysis.

3 Four points: trivial warm-up

The square has two chords, X_{13} and X_{24}. The ansatz

$$B = \frac{a_{X_{13}}}{X_{13}} + \frac{a_{X_{24}}}{X_{24}}$$

has 2 coefficients, both *local* (each chord by itself is a triangulation). There are no non-local multisets. A single zero suffices to recover amplitude unitarity (equal residues on both channels) but no locality question arises; we move on to $n = 5$.

4 Five points: full proof

4.1 The pentagon's chords, enclosed vertices, and crossings

The five chords of the pentagon, with their enclosed-vertex sets:

Chord	X_{13}	X_{14}	X_{24}	X_{25}	X_{35}
$E(\cdot)$	$\{2\}$	$\{5\}$	$\{3\}$	$\{1\}$	$\{4\}$

Every chord has class 1 at $n = 5$ (class-2 chords first appear at $n = 6$).

There are $\binom{5}{2} = 10$ unordered pairs of distinct chords; 5 cross, 5 share an endpoint. The 5 crossings (by Lemma 1) are:

$$X_{13} \times X_{24}, \quad X_{13} \times X_{25}, \quad X_{14} \times X_{25}, \quad X_{14} \times X_{35}, \quad X_{24} \times X_{35}.$$

The 5 non-crossing pairs are exactly the 5 fan triangulations.

4.2 The 15-coefficient ansatz

With $n - 3 = 2$ chord factors per monomial:

$$B = \sum_{1 \le i \le j \le 5} \frac{a_{ij}}{X_i X_j}, \quad \text{15 coefficients.}$$

Of these 15:

- 5 *tree triples* (fan triangulations);
- 5 *double poles* $a_{X_i X_i}/X_i^2$;
- 5 *crossing pairs* $a_{X_i X_j}/(X_i X_j)$ with $X_i \times X_j$ crossing.

The 10 non-local coefficients are the 5 double poles and the 5 crossing pairs.

4.3 Deriving the five cyclic hidden zeros

Each $\mathcal{Z}_r$ is defined by $c_{r,r+2} = c_{r,r+3} = 0$ (two constraints at $n = 5$). We derive each by applying the mesh identity (1).

$\mathcal{Z}_1$:

$$c_{1,3} = X_{14} + X_{23} - X_{13} - X_{24} = X_{14} - X_{13} - X_{24} = 0,$$
$$c_{1,4} = X_{15} + X_{24} - X_{14} - X_{25} = X_{24} - X_{14} - X_{25} = 0.$$

(Used: $X_{23} = X_{15} = 0$.) Solving for X_{24}, X_{25}:

$$\mathcal{Z}_1 : \qquad X_{24} = X_{14} - X_{13}, \qquad X_{25} = -X_{13}. \tag{2}$$

The chord X_{13} appears on both right-hand sides: it is the **special variable**. Both substituted chords are chords from the enclosed vertex 2 of X_{13}.

$\mathcal{Z}_2$ through $\mathcal{Z}_5$: By cyclic rotation of the derivation, or equivalently by direct application of (1):

$$\mathcal{Z}_2 : X_{35} = X_{25} - X_{24}, \quad X_{13} = -X_{24}; \quad \text{special } X_{24}. \tag{3}$$
$$\mathcal{Z}_3 : X_{14} = X_{13} - X_{35}, \quad X_{24} = -X_{35}; \quad \text{special } X_{35}. \tag{4}$$
$$\mathcal{Z}_4 : X_{25} = X_{24} - X_{14}, \quad X_{35} = -X_{14}; \quad \text{special } X_{14}. \tag{5}$$
$$\mathcal{Z}_5 : X_{13} = X_{35} - X_{25}, \quad X_{14} = -X_{25}; \quad \text{special } X_{25}. \tag{6}$$

4.4 Canonical vocabulary for every zone

Definition 2 (Canonical vocabulary)**.** At $\mathcal{Z}_r$ in canonical form:

- **Special variable** X_*: the chord in the RHS of every substitution. It is the short diagonal X_{rr+2} whose enclosed vertex is $r + 1$.
- **Substituted chords** S_r: the chords on the LHS. At $\mathcal{Z}_r$ they are the chords originating at the enclosed vertex $r + 1$.
- A substituted chord X_s is **non-bare** if its substitution has form $X_s = X_f - X_*$ with X_f a genuinely free (non-zero) chord; X_f is its **companion**, denoted f_s. It is **bare** if $X_s = -X_*$ (companion 0).
- **Free set** F_r: chords that are neither substituted nor special.

At $n = 5$, every zone has $|F_r| = 2$, one non-bare substitution with its companion in F_r, and one bare substitution. Tabulating:

Zone	Special X_*	Non-bare X_s	Companion f_s	Bare
$\mathcal{Z}_1$	X_{13}	X_{24}	X_{14}	X_{25}
$\mathcal{Z}_2$	X_{24}	X_{35}	X_{25}	X_{13}
$\mathcal{Z}_3$	X_{35}	X_{14}	X_{13}	X_{24}
$\mathcal{Z}_4$	X_{14}	X_{25}	X_{24}	X_{35}
$\mathcal{Z}_5$	X_{25}	X_{13}	X_{35}	X_{14}

4.5 The kill technique at $\mathcal{Z}_1$: two layers

For $\mathcal{Z}_1$: $X_* = X_{13}$, non-bare $X_{24} = X_{14} - X_{13}$ with companion $f = X_{14}$, bare $X_{25} = -X_{13}$, free set $F_1 = \{X_{14}, X_{35}\}$.

After substitution, $B|_{\mathcal{Z}_1}$ is a rational function of the three variables X_{13}, X_{14}, X_{35}. Since $B|_{\mathcal{Z}_1} = 0$ identically, its Laurent expansion in $X_* = X_{13}$ vanishes order by order, with coefficients that are rational functions of the two "free" variables.

The key Laurent expansions:

$$\frac{1}{X_{24}} = \frac{1}{X_{14} - X_{13}} = -\frac{1}{X_{13}} - \frac{X_{14}}{X_{13}^2} - \frac{X_{14}^2}{X_{13}^3} - \cdots, \tag{7}$$

$$\frac{1}{X_{25}} = \frac{1}{-X_{13}} = -\frac{1}{X_{13}} \qquad \text{(exact, no tail).} \tag{8}$$

Layer 1: Order X_{13}^0, the limit $X_{13} \to \infty$. As $X_{13} \to \infty$ with X_{14}, X_{35} held fixed, we have $1/X_{24} \to 0$ and $1/X_{25} \to 0$. So every monomial of B containing X_{13}, X_{24}, or X_{25} vanishes in the limit. The only surviving monomials are those with both factors in F_1:

$$\lim_{X_{13}\to\infty} B|_{\mathcal{Z}_1} = \frac{a_{X_{14}X_{14}}}{X_{14}^2} + \frac{a_{X_{35}X_{35}}}{X_{35}^2} + \frac{a_{X_{14}X_{35}}}{X_{14}X_{35}}.$$

The three rational functions $1/X_{14}^2,\ 1/X_{35}^2,\ 1/(X_{14}X_{35})$ are linearly independent, so each coefficient must vanish:

$$\boxed{a_{X_{14}X_{14}} = a_{X_{35}X_{35}} = a_{X_{14}X_{35}} = 0.} \quad \text{(3 order-0 kills at } \mathcal{Z}_1\text{)}$$

Layer 2: Order $1/X_{13}^2$. To verify the single-term kill at this order, we tabulate the contribution of each of the 15 monomials of B to the $1/X_{13}^2$ coefficient of $B|_{\mathcal{Z}_1}$:

#	Monomial	Contribution to $(B\|_{\mathcal{Z}_1})_{1/X_{13}^2}$
1	$a_{X_{13}X_{13}}/X_{13}^2$	$+a_{X_{13}X_{13}}$
2	$a_{X_{13}X_{14}}/(X_{13}X_{14})$	0 (contributes at $1/X_{13}$)
3	$a_{X_{13}X_{24}}/(X_{13}X_{24})$	$-a_{X_{13}X_{24}}$
4	$a_{X_{13}X_{25}}/(X_{13}X_{25})$	$-a_{X_{13}X_{25}}$
5	$a_{X_{13}X_{35}}/(X_{13}X_{35})$	0
6	$a_{X_{14}X_{14}}/X_{14}^2$	0
7	$a_{X_{14}X_{24}}/(X_{14}X_{24})$	$-a_{X_{14}X_{24}}$
8	$a_{X_{14}X_{25}}/(X_{14}X_{25})$	0
9	$a_{X_{14}X_{35}}/(X_{14}X_{35})$	0
10	$a_{X_{24}X_{24}}/X_{24}^2$	$+a_{X_{24}X_{24}}$
11	$a_{X_{24}X_{25}}/(X_{24}X_{25})$	$+a_{X_{24}X_{25}}$
12	$a_{X_{24}X_{35}}/(X_{24}X_{35})$	$\boldsymbol{-a_{X_{24}X_{35}} \cdot X_{14}/X_{35}}$
13	$a_{X_{25}X_{25}}/X_{25}^2$	$+a_{X_{25}X_{25}}$
14	$a_{X_{25}X_{35}}/(X_{25}X_{35})$	0
15	$a_{X_{35}X_{35}}/X_{35}^2$	0

Spot-checks. Row 7: $1/(X_{14}X_{24}) = (1/X_{14})(-1/X_{13} - X_{14}/X_{13}^2 - \cdots)$; $1/X_{13}^2$ piece is $-1/X_{13}^2$, coefficient -1. Row 10: $1/X_{24}^2 = 1/X_{13}^2(1 - X_{14}/X_{13})^{-2} = 1/X_{13}^2 + 2X_{14}/X_{13}^3 + \cdots$, coefficient $+1$. Row 12: $1/(X_{24}X_{35}) = (1/X_{35})(-1/X_{13} - X_{14}/X_{13}^2 - \cdots)$; $1/X_{13}^2$ piece is $-X_{14}/(X_{35}X_{13}^2)$, coefficient $-X_{14}/X_{35}$.

Summing column 3:

$$(B|_{\mathcal{Z}_1})_{1/X_{13}^2} = \underbrace{[a_{X_{13}X_{13}} - a_{X_{13}X_{24}} - a_{X_{13}X_{25}} - a_{X_{14}X_{24}} + a_{X_{24}X_{24}} + a_{X_{24}X_{25}} + a_{X_{25}X_{25}}]}_{K} - a_{X_{24}X_{35}} \cdot \frac{X_{14}}{X_{35}}.$$

Setting this to zero as a rational function of two independent variables X_{14}, X_{35}: the functions 1 and X_{14}/X_{35} are linearly independent, so both coefficients vanish separately. This gives the multi-term equation $K = 0$ and, more importantly,

$$\boxed{a_{X_{24}X_{35}} = 0.} \quad \text{(1 order-2 kill at } \mathcal{Z}_1\text{)}$$

Summary for $\mathcal{Z}_1$. Four single-coefficient kills: three order-0 (all multisets in F_1) and one order-2 (the crossing pair $\{X_s, X_b\}$ where $X_s = X_{24}$ is the non-bare substituted chord and $X_b = X_{35}$ is the free chord $\neq f_s = X_{14}$).

4.6 The companion-isolation rule at $n = 5$

The order-2 mechanism generalizes cleanly. Examining row 12: the monomial of B produced the fingerprint X_{14}/X_{35}, a non-constant rational function of the two free variables. All other monomials produced only constants at order $1/X_{13}^2$. The mechanism is:

Lemma 2 (Companion-isolation, $n = 5$)**.** *Let $\mathcal{Z}_r$ have non-bare substitution $X_s = X_f - X_*$ with companion $X_f \in F_r$. For each free chord $X_b \in F_r \setminus \{X_f\}$, the monomial X_f/X_b at order $1/X_*^2$ has a unique source in $B|_{\mathcal{Z}_r}$: $a_{X_s X_b}/(X_s X_b)$. Consequently*

$$a_{X_s X_b} = 0 \quad \textit{for every } X_b \in F_r \setminus \{X_f\}.$$

Proof. The Laurent expansion of $1/X_s = 1/(X_f - X_*)$ contributes $-X_f/X_*^2$ at order $1/X_*^2$. Multiplying by a free factor $1/X_b$ (with $X_b \neq X_f$) gives the monomial $-X_f/(X_*^2 X_b)$.

No other monomial of B produces X_f/X_b at order $1/X_*^2$:

- Monomials with no substituted factor: contribute at order X_*^0.
- Monomials with only the bare substituted chord (and frees): the bare $1/X_{\text{bare}} = -1/X_*$ has no Laurent tail, so contributions at $1/X_*^2$ or higher have no free-variable dependence.
- Monomials with X_* as a factor and a free X_b: $1/(X_* X_b)$ contributes at order $1/X_*$, not $1/X_*^2$.
- Monomials with both X_* and X_s: $1/(X_* X_s) = (1/X_*)(-1/X_* + \cdots)$ contributes $-1/X_*^2$ at leading order (a constant).
- Monomials with X_s twice: $1/X_s^2$ has leading piece $1/X_*^2$ (constant in free variables).
- Monomials with $X_s\, X_{\text{bare}}$: $1/(X_s X_{\text{bare}}) = -(1/X_*) \cdot 1/X_s$, contributes $+1/X_*^2$ at leading order (constant).
- Monomials with X_s and a free $X_b = X_f$: contributes $-X_f/(X_*^2 X_f) = -1/X_*^2$, a constant — the X_f "cancels" with $1/X_f$ in the denominator, losing its distinctness.

Only $a_{X_s X_b}/(X_s X_b)$ with $X_b \neq X_f$ produces the non-constant fingerprint X_f/X_b. Hence the kill. □

At $n = 5$, $|F_r| - 1 = 1$, so this gives exactly 1 order-2 kill per zone.

4.7 Cycling: all five zones and the union

By cyclic symmetry, each $\mathcal{Z}_r$ produces 3 order-0 kills (multisets in F_r) and 1 order-2 kill (the Lemma 2 product). Tabulating:

Zone	Order-0 kills (multisets in F_r)	Order-2 kill
$\mathcal{Z}_1$	$a_{X_{14}X_{14}}, a_{X_{35}X_{35}}, a_{X_{14}X_{35}}$	$a_{X_{24}X_{35}}$
$\mathcal{Z}_2$	$a_{X_{14}X_{14}}, a_{X_{25}X_{25}}, a_{X_{14}X_{25}}$	$a_{X_{35}X_{14}}$
$\mathcal{Z}_3$	$a_{X_{13}X_{13}}, a_{X_{25}X_{25}}, a_{X_{13}X_{25}}$	$a_{X_{14}X_{25}}$
$\mathcal{Z}_4$	$a_{X_{13}X_{13}}, a_{X_{24}X_{24}}, a_{X_{13}X_{24}}$	$a_{X_{25}X_{13}}$
$\mathcal{Z}_5$	$a_{X_{24}X_{24}}, a_{X_{35}X_{35}}, a_{X_{24}X_{35}}$	$a_{X_{13}X_{24}}$

Union check. Every non-local coefficient appears as a kill:

- Double poles: $a_{X_{13}^2}$ (at $\mathcal{Z}_3, \mathcal{Z}_4$), $a_{X_{14}^2}$ (at $\mathcal{Z}_1, \mathcal{Z}_2$), $a_{X_{24}^2}$ (at $\mathcal{Z}_4, \mathcal{Z}_5$), $a_{X_{25}^2}$ (at $\mathcal{Z}_2, \mathcal{Z}_3$), $a_{X_{35}^2}$ (at $\mathcal{Z}_1, \mathcal{Z}_5$).
- Crossing pairs: $a_{X_{13}X_{24}}$ (at $\mathcal{Z}_4$ order 0, $\mathcal{Z}_5$ order 2), $a_{X_{13}X_{25}}$ (at $\mathcal{Z}_3$ order 0, $\mathcal{Z}_4$ order 2), $a_{X_{14}X_{25}}$ (at $\mathcal{Z}_2$ order 0, $\mathcal{Z}_3$ order 2), $a_{X_{14}X_{35}}$ (at $\mathcal{Z}_1$ order 0, $\mathcal{Z}_2$ order 2), $a_{X_{24}X_{35}}$ (at $\mathcal{Z}_1$ order 2, $\mathcal{Z}_5$ order 0).

All 10 non-local coefficients killed. Every crossing pair is killed by exactly two cyclically adjacent zones, once at order 0 and once at order 2.

4.8 Trees are not killed at $n = 5$

At each zone, the 3 order-0 kills are multisets $\subseteq F_r$ and the 1 order-2 kill uses the non-bare-substituted chord. A tree triple $\{X_{ab}, X_{bc}\}$ (sharing vertex b) is never in any F_r because for each r, the fan from a specific vertex is broken by the substitution structure. A direct check of each tree against the 5 free sets confirms this.

Theorem 2 (Locality at five points)**.** *The five hidden one-zeros of the pentagon force every non-local coefficient of the general rational ansatz to zero. The* 5 *surviving coefficients are exactly the tree triangulations.*

5 Six points: full proof

At $n = 6$ the hexagon has two classes of chords (short and long diagonals), and the kill structure acquires two "layers". The argument follows the $n = 5$ template but with richer combinatorics, and we need an orbit-by-orbit analysis to verify that every one of the 151 non-local multisets is killed.

5.1 Setup

The hexagon's 9 chords split as:

Short diagonals (class 1, enclose 1): $X_{13}, X_{24}, X_{35}, X_{46}, X_{15}, X_{26}$
Long diagonals (class 2, enclose 2): X_{14}, X_{25}, X_{36}

We use cyclic labels:

$$x_r = X_{rr+2} \text{ (short, } r = 1, \ldots, 6 \bmod 6), \qquad y_r = X_{rr+3} \ (r = 1, 2, 3 \bmod 3).$$

Enclosed-vertex sets: $E(x_r) = \{r + 1\} \pmod 6$, $E(y_r) = \{r + 1, r + 2\} \pmod 6$.

The ansatz has $\binom{9+2}{3} = 165$ coefficients. Of these, $14 = C_4$ are tree triples (the hexagon triangulations), leaving 151 non-local.

5.2 The six zones in canonical form

Each $\mathcal{Z}_r$ imposes 3 linear constraints $c_{r,r+2} = c_{r,r+3} = c_{r,r+4} = 0$. We derive $\mathcal{Z}_1$ explicitly from (1):

$$\begin{aligned} c_{1,3} &= X_{14} + X_{23} - X_{13} - X_{24} = y_1 - x_1 - x_2 = 0, \\ c_{1,4} &= X_{15} + X_{24} - X_{14} - X_{25} = x_5 + x_2 - y_1 - y_2 = 0, \\ c_{1,5} &= X_{16} + X_{25} - X_{15} - X_{26} = 0 + y_2 - x_5 - x_6 = 0. \end{aligned}$$

(Used: $X_{23} = X_{16} = 0$.) Solving for the three chords originating at vertex 2 (the enclosed vertex of x_1):

$$\mathcal{Z}_1 : \qquad x_2 = y_1 - x_1, \quad y_2 = x_5 - x_1, \quad x_6 = -x_1.$$

Special: x_1. Two non-bare substitutions (x_2, y_2 with companions y_1, x_5) and one bare (x_6). Free set: $F_1 = \{x_3, x_4, x_5, y_1, y_3\}$.

The other five zones follow by cyclic rotation ($\sigma : r \mapsto r + 1$ on x-indices mod 6, on y-indices mod 3, and on zone indices mod 6):

Zone	Substitutions	Non-bare + comp.	Bare
$\mathcal{Z}_1$	$x_2 = y_1 - x_1,\ y_2 = x_5 - x_1,\ x_6 = -x_1$	$x_2\,(y_1),\ y_2\,(x_5)$	x_6
$\mathcal{Z}_2$	$x_3 = y_2 - x_2,\ y_3 = x_6 - x_2,\ x_1 = -x_2$	$x_3\,(y_2),\ y_3\,(x_6)$	x_1
$\mathcal{Z}_3$	$x_4 = y_3 - x_3,\ y_1 = x_1 - x_3,\ x_2 = -x_3$	$x_4\,(y_3),\ y_1\,(x_1)$	x_2
$\mathcal{Z}_4$	$x_5 = y_1 - x_4,\ y_2 = x_2 - x_4,\ x_3 = -x_4$	$x_5\,(y_1),\ y_2\,(x_2)$	x_3
$\mathcal{Z}_5$	$x_6 = y_2 - x_5,\ y_3 = x_3 - x_5,\ x_4 = -x_5$	$x_6\,(y_2),\ y_3\,(x_3)$	x_4
$\mathcal{Z}_6$	$x_1 = y_3 - x_6,\ y_1 = x_4 - x_6,\ x_5 = -x_6$	$x_1\,(y_3),\ y_1\,(x_4)$	x_5

Free sets:

$$\begin{aligned} F_1 &= \{x_3, x_4, x_5, y_1, y_3\}, \\ F_2 &= \{x_4, x_5, x_6, y_1, y_2\}, \\ F_3 &= \{x_1, x_5, x_6, y_2, y_3\}, \\ F_4 &= \{x_1, x_2, x_6, y_1, y_3\}, \\ F_5 &= \{x_1, x_2, x_3, y_1, y_2\}, \\ F_6 &= \{x_2, x_3, x_4, y_2, y_3\}. \end{aligned} \tag{9}$$

Structural observations. Each F_r has 3 shorts and 2 longs. The short parts are the six cyclic 3-windows $\{x_{r+2}, x_{r+3}, x_{r+4}\}$ (mod 6). The long parts are $\{y_1, y_2, y_3\} \setminus \{y_{r+1}\}$ (mod 3), i.e., each F_r omits exactly one y — the non-bare-long-substituted one.

5.3 The two kill layers at $\mathcal{Z}_1$

For $\mathcal{Z}_1$: $X_* = x_1$. Laurent expansions of the substituted chords:

$$\frac{1}{x_2} = \frac{1}{y_1 - x_1} = -\frac{1}{x_1} - \frac{y_1}{x_1^2} - \frac{y_1^2}{x_1^3} - \cdots, \tag{10}$$

$$\frac{1}{y_2} = \frac{1}{x_5 - x_1} = -\frac{1}{x_1} - \frac{x_5}{x_1^2} - \frac{x_5^2}{x_1^3} - \cdots, \tag{11}$$

$$\frac{1}{x_6} = -\frac{1}{x_1} \qquad \text{(exact)}. \tag{12}$$

Layer 1: Order x_1^0. (35 kills at $\mathcal{Z}_1$.) As $x_1 \to \infty$ with the five free variables $\{x_3, x_4, x_5, y_1, y_3\}$ fixed, every monomial of B containing x_1, x_2, y_2, or x_6 vanishes in the limit. The surviving monomials are those with all three indices in F_1:

$$\lim_{x_1 \to \infty} B|_{\mathcal{Z}_1} = \sum_{(X_a, X_b, X_c) \subseteq F_1} \frac{a_{X_a X_b X_c}}{X_a X_b X_c}.$$

The number of multisets of size 3 from a 5-element set is $\binom{5+2}{3} = 35$.

Lemma 3 (Monomial linear independence)**.** *Let $X_1, \ldots, X_m$ be m independent variables. The $\binom{m+k-1}{k}$ rational functions $\{1/(X_{i_1} \cdots X_{i_k}) : \text{multisets of size } k\}$ are linearly independent over $\mathbb{Q}$.*

Proof. Suppose $\sum c_{(i_1,\ldots,i_k)} \prod_j 1/X_{i_j} = 0$. Multiply through by $\prod_i X_i^k$ (clearing denominators). Each term becomes a distinct monomial $\prod_i X_i^{k-m_i}$, where m_i is the multiplicity of X_i in the multiset; distinct multisets give distinct exponent tuples. Monomials in independent variables are linearly independent, so every $c = 0$. □

Hence every $a_{X_a X_b X_c}$ with $\{X_a, X_b, X_c\} \subseteq F_1$ vanishes. **35 Layer-1 kills at $\mathcal{Z}_1$.**

Layer 2: Order $1/x_1^2$. (20 new kills at $\mathcal{Z}_1$.) The analysis parallels $n = 5$, except each monomial of B has 3 factors, so the "other two factors" of a single-non-bare term must be a 2-multiset of free chords.

Lemma 4 (Companion-isolation at $n = 6$)**.** *At $\mathcal{Z}_r$, let $X_s = X_f - X_*$ be a non-bare substitution (companion $X_f \in F_r$). For each multiset (X_a, X_b) of size 2 from $F_r \setminus \{X_f\}$, the monomial $X_f/(X_a X_b)$ at order $1/X_*^2$ in $B|_{\mathcal{Z}_r}$ has a unique source, namely $a_{X_s X_a X_b}/(X_s X_a X_b)$. Hence*

$$a_{X_s X_a X_b} = 0 \quad \textit{for every multiset } (X_a, X_b) \subseteq F_r \setminus \{X_f\}.$$

Proof sketch. The Laurent tail $-X_f/X_*^2$ of $1/X_s$, multiplied by $1/(X_a X_b)$ with $\{X_a, X_b\} \subseteq F_r \setminus \{X_f\}$, gives the non-constant fingerprint $-X_f/(X_*^2 X_a X_b)$ with unique free-variable content $X_f/(X_a X_b)$.

To check no other monomial of B produces this fingerprint at order $1/X_*^2$, we enumerate possibilities (analogous to the $n = 5$ proof): any other source must have either a different non-bare-substituted chord (different companion), or produce X_f from a different algebraic source; each such source is ruled out. Crucially, if X_a or X_b equals X_f, the monomial degenerates to a constant and loses distinctness — this is why we require $X_b \neq X_f$. □

Applied at $\mathcal{Z}_1$:

- Non-bare x_2 (comp y_1): multisets of size 2 from $F_1 \setminus \{y_1\} = \{x_3, x_4, x_5, y_3\}$: $\binom{4+1}{2} = 10$ kills.
- Non-bare y_2 (comp x_5): multisets of size 2 from $F_1 \setminus \{x_5\} = \{x_3, x_4, y_1, y_3\}$: another 10 kills.

Total Layer-2 kills at $\mathcal{Z}_1$: **20**.

Total at $\mathcal{Z}_1$: 55 single-term kills (35 order 0 + 20 order 2). By cyclic symmetry, each $\mathcal{Z}_r$ gives 55 kills of the same structure.

5.4 The Layer-1 coverage count: 129

How many distinct non-local coefficients are covered by Layer 1 across all 6 zones? Equivalently, how many non-local multisets lie in at least one F_r?

We enumerate the complement — multisets *not* in any F_r — by chord-class composition.

Type (3, 0): all-short multisets. A multiset $\{x_a, x_b, x_c\}$ (with possible repetition) lies in F_r iff its index set $\{a, b, c\}$ lies in some cyclic 3-window $\{r+2, r+3, r+4\}$. There are 6 such windows; a distinct-index triple is covered iff it equals such a window.

Enumerate:

- Squared multisets x_a^3 (6 total): each index is in many windows. All covered.
- Mixed double-single $x_a^2 x_b$, $a \neq b$ (30 total): covered iff the pair $\{a, b\}$ fits in a 3-window, i.e., cyclic distance < 3. Pairs with cyclic distance exactly 3: $\{1, 4\}, \{2, 5\}, \{3, 6\}$ — 3 pairs, each giving 2 multisets. **6 failures.**
- Distinct triples (20 total): covered iff a 3-window matches. There are 6 such "consecutive" triples. Among the remaining 14 non-consecutive triples, 2 are trees ($\{1, 3, 5\}$ and $\{2, 4, 6\}$, the two "zigzag" triangulations), leaving **12 non-tree failures.**

Type-(3,0) Layer-1 failures: $6 + 12 = 18$.

Type (0, 3): all-long multisets (10 total). A multiset $\{y_a, y_b, y_c\}$ lies in F_r iff its distinct indices lie in $\{1, 2, 3\} \setminus \{r+1 \bmod 3\}$. Each F_r omits exactly one y; across 6 zones, every y is present in 4 zones. A multiset using only 2 distinct y's is covered (pick the zone omitting the third). A multiset using all 3 distinct y's is *not* covered by any zone.

Single non-tree failure: $\{y_1, y_2, y_3\}$.

Type (1, 2): one short, two long (36 total, 0 trees). A multiset $\{x_a, y_c, y_d\}$ is covered iff y_c, y_d both lie in some F_r's y-pair AND x_a lies in that zone's x-part.

For each y-pair $\{y_i, y_j\}$ ($i \neq j$) there are two zones whose y-part is $\{y_i, y_j\}$; their x-parts are cyclic shifts differing by 3, and their union is $\{x_1, \ldots, x_6\}$ (all 6 shorts). So every $\{x_a, y_i, y_j\}$ with $i \neq j$ is covered.

For y_i^2 (two copies of same y), $y_i \in F_r$ for 4 zones; their x-parts union to all shorts. So every $\{x_a, y_i^2\}$ is covered.

Zero type-(1,2) failures.

Type (2, 1): two short, one long. We enumerate with $y = y_1$ fixed; the y_2 and y_3 slices follow by symmetry.

With $y_1 \in F_r$ (zones $r \in \{1, 2, 4, 5\}$), the available x-windows are $\{3, 4, 5\}, \{4, 5, 6\}, \{1, 2, 6\}, \{1, 2, 3\}$.

A multiset $\{x_a, x_b, y_1\}$ is Layer-1 covered iff the pair $\{a, b\}$ (as a multiset) lies in one of these 4 windows. Squared-x multisets $\{x_a^2, y_1\}$: the union of x-parts above is $\{1, \ldots, 6\}$ (all), so every squared-x multiset is covered.

Distinct-pair multisets $\{x_a, x_b, y_1\}$ with $a < b$: enumerate 15 pairs and check which fit in one of the 4 windows. A quick case analysis yields 5 failures: pairs $\{1, 4\}, \{1, 5\}, \{2, 4\}, \{2, 5\}, \{3, 6\}$.

Among these 5: the first four are *trees* ($\{x_1, x_4, y_1\} = 147$, etc., all fans), so they're supposed to survive. The remaining 1 non-tree failure is $\{x_3, x_6, y_1\}$.

By cyclic symmetry on y: 1 failure with y_2, 1 with y_3. **Total type-(2,1) failures: 3.**

Total Layer-1 failures.

$$18 + 1 + 0 + 3 = \mathbf{22} \text{ non-local Layer-1 failures; Layer-1 covered: } 151 - 22 = \mathbf{129}.$$

5.5 Blocking-set property: no tree in any F_r

Lemma 5 (Blocking set)**.** *No hexagon triangulation is entirely contained in any F_r.*

Proof. By cyclic symmetry, suffices to check $F_1 = \{x_3, x_4, x_5, y_1, y_3\}$ against the 14 tree triples:

Tree	Chord form	Excluded chord	Tree	
135	$\{x_1, x_3, x_5\}$	x_1	258	$\{x_2, x_5, y_2\}$: x_2, y_2
139	$\{x_1, x_3, y_3\}$	x_1	268	$\{x_2, x_6, y_2\}$: x_2, x_6, y_2
147	$\{x_1, x_4, y_1\}$	x_1	358	$\{x_3, x_5, y_2\}$: y_2
149	$\{x_1, x_4, y_3\}$	x_1	368	$\{x_3, x_6, y_2\}$: x_6, y_2
157	$\{x_1, x_5, y_1\}$	x_1	369	$\{x_3, x_6, y_3\}$: x_6
246	$\{x_2, x_4, x_6\}$	x_2, x_6	469	$\{x_4, x_6, y_3\}$: x_6
247	$\{x_2, x_4, y_1\}$	x_2		
257	$\{x_2, x_5, y_1\}$	x_2		

Every tree contains at least one chord outside F_1. By cyclic rotation, the same holds for every F_r. □

Corollary 1. *The Layer-1 argument at $\mathcal{Z}_r$ only kills non-tree multisets. No tree is accidentally killed.*

5.6 Layer-2 analysis for the 22 remaining multisets: orbit structure

The 22 Layer-1 failures fall into 5 cyclic-symmetry orbits:

Orbit label	Size	Representative	Size×(per orbit)	Killing zone (L2)
(a) Squared distance-3 pairs	6	$x_1^2 x_4$	6	$\mathcal{Z}_6$
(b) Non-consec triples, sub-orbit α	6	$\{x_1, x_2, x_4\}$	6	$\mathcal{Z}_6$
(c) Non-consec triples, sub-orbit β	6	$\{x_1, x_2, x_5\}$	6	$\mathcal{Z}_4$
(d) Crossed y-fork	3	$\{x_3, x_6, y_1\}$	3	$\mathcal{Z}_2$
(e) All-three-y	1	$\{y_1, y_2, y_3\}$	1	$\mathcal{Z}_1$
Total			**22**	

Orbit verification. Under σ (shift by 1 on x-indices mod 6 and y-indices mod 3 simultaneously):

- Orbit (a): $x_1^2 x_4 \to x_2^2 x_5 \to x_3^2 x_6 \to x_4^2 x_1 \to x_5^2 x_2 \to x_6^2 x_3 \to x_1^2 x_4$. Size 6.
- Orbit (b): $\{1, 2, 4\} \to \{2, 3, 5\} \to \{3, 4, 6\} \to \{1, 4, 5\} \to \{2, 5, 6\} \to \{1, 3, 6\} \to \{1, 2, 4\}$. Size 6.
- Orbit (c): $\{1, 2, 5\} \to \{2, 3, 6\} \to \{1, 3, 4\} \to \{2, 4, 5\} \to \{3, 5, 6\} \to \{1, 4, 6\} \to \{1, 2, 5\}$. Size 6.
- Orbit (d): $\{x_3, x_6, y_1\} \to \{x_4, x_1, y_2\} \to \{x_5, x_2, y_3\} \to \{x_3, x_6, y_1\}$ (size 3; σ^3 fixes y-index mod 3 and swaps $x_i \leftrightarrow x_{i+3}$, leaving the unordered pair $\{3, 6\}$ invariant).
- Orbit (e): $\{y_1, y_2, y_3\}$ is fixed by σ (the y-indices permute among themselves). Size 1.

We now verify one representative of each orbit. The remaining members follow by applying σ to both the multiset and the zone.

5.7 Verification of orbit representatives

Orbit (a): $x_1^2 x_4 = a_{x_1 x_1 x_4}$ at $\mathcal{Z}_6$ Layer 2

$\mathcal{Z}_6$ substitutions: $x_1 = y_3 - x_6$ (non-bare, comp y_3), $y_1 = x_4 - x_6$ (non-bare, comp x_4), $x_5 = -x_6$ (bare). Special: x_6. Free: $F_6 = \{x_2, x_3, x_4, y_2, y_3\}$.

The target monomial on $\mathcal{Z}_6$:

$$\frac{a_{x_1^2 x_4}}{x_1^2\, x_4} = \frac{a_{x_1^2 x_4}}{(y_3 - x_6)^2\, x_4}.$$

Expand $1/(y_3 - x_6)^2$:

$$\frac{1}{(y_3 - x_6)^2} = \frac{1}{x_6^2}\left(1 + \frac{2y_3}{x_6} + \frac{3y_3^2}{x_6^2} + \cdots\right) = \frac{1}{x_6^2} + \frac{2y_3}{x_6^3} + \frac{3y_3^2}{x_6^4} + \cdots$$

At order $1/x_6^3$: the coefficient is $2\, a_{x_1^2 x_4}\, y_3/x_4$.

Fingerprint: y_3/x_4 at order $1/x_6^3$.

Unique source check. On $\mathcal{Z}_6$, the only source of y_3 in a Laurent numerator is $1/x_1 = -1/x_6 - y_3/x_6^2 - y_3^2/x_6^3 - \cdots$. (The other non-bare $1/y_1$ produces powers of x_4, not y_3; the bare $1/x_5 = -1/x_6$ has no tail.)

So any monomial of B producing y_3 at any order must have x_1 as a factor. Cases:

- **Monomial with x_1 once:** $a_{x_1 X_a X_b}/(x_1 X_a X_b)$. At order $1/x_6^3$: $1/x_1$ contributes $-y_3^2/x_6^3$; fingerprint is $y_3^2/(X_a X_b)$, not y_3/x_4.
- **Monomial with x_1 twice:** $a_{x_1^2 X_c}/(x_1^2 X_c)$. At order $1/x_6^3$: $1/x_1^2$ contributes $2y_3/x_6^3$; times $1/X_c$ gives $2y_3/(x_6^3 X_c)$. Fingerprint is y_3/X_c.
- **Monomial with x_1 three times:** $a_{x_1^3}/x_1^3$. At order $1/x_6^3$: $1/x_1^3 = 1/(y_3 - x_6)^3$, leading term $-1/x_6^3$ (constant in free variables), no y_3 at this order.

So the only candidates producing the fingerprint y_3/x_4 at $1/x_6^3$ are $a_{x_1^2 X_c}$ with $X_c = x_4$. The single coefficient $a_{x_1^2 x_4}$ is isolated.

Additionally, we must check that $X_c = x_4$ doesn't degenerate (i.e., that $x_4 \neq y_3$, the companion of x_1). Indeed $x_4 \neq y_3$, so the fingerprint is non-constant and the kill is clean.

$$\boxed{a_{x_1^2 x_4} = 0.}$$

By cyclic rotation, the entire orbit (a) (6 multisets) is killed: $a_{x_r^2 x_{r+3}}$ and $a_{x_r x_{r+3}^2}$ at $\mathcal{Z}_{r+5}$ for $r = 1, \ldots, 6$ (indices mod 6).

Orbit (b): $\{x_1, x_2, x_4\} = a_{x_1 x_2 x_4}$ at $\mathcal{Z}_6$ Layer 2

The target on $\mathcal{Z}_6$:

$$\frac{a_{x_1 x_2 x_4}}{x_1\, x_2\, x_4} = \frac{a_{x_1 x_2 x_4}}{(y_3 - x_6)\, x_2\, x_4}.$$

At order $1/x_6^2$: $1/x_1$ contributes $-y_3/x_6^2$, so the coefficient is $-a_{x_1 x_2 x_4}\, y_3/(x_2\, x_4)$.

Fingerprint: $y_3/(x_2\, x_4)$ at order $1/x_6^2$.

Unique source check. Again, y_3 comes only from $1/x_1$'s tail. For a single-x_1 monomial $a_{x_1 X_a X_b}/(x_1 X_a X_b)$ at order $1/x_6^2$, the $1/x_1$ piece $-y_3/x_6^2$ pairs with $1/(X_a X_b)$. The monomial $y_3/(X_a X_b)$ uniquely identifies $\{X_a, X_b\}$ as a multiset, provided $X_a, X_b \neq y_3$ (else degenerate).

For multisets with x_1 appearing more than once, the Laurent contribution at order $1/x_6^2$ is $1/x_1^2 = 1/x_6^2 + O(1/x_6^3)$ — pure constant at leading order, no y_3 yet. So no contribution to the $y_3/(x_2\, x_4)$ fingerprint.

Hence $a_{x_1 x_2 x_4}$ is uniquely isolated. Since $\{x_2, x_4\} \subseteq F_6 \setminus \{y_3\} = \{x_2, x_3, x_4, y_2\}$, the fingerprint is non-constant.

$$\boxed{a_{x_1 x_2 x_4} = 0.}$$

Cyclic rotation kills the full 6-orbit.

Orbit (c): $\{x_1, x_2, x_5\} = a_{x_1x_2x_5}$ at $\mathcal{Z}_4$ Layer 2

$\mathcal{Z}_4$ substitutions: $x_5 = y_1 - x_4$ (non-bare, comp y_1), $y_2 = x_2 - x_4$ (non-bare, comp x_2), $x_3 = -x_4$ (bare). Special: x_4. Free: $F_4 = \{x_1, x_2, x_6, y_1, y_3\}$.

The target on $\mathcal{Z}_4$:

$$\frac{a_{x_1x_2x_5}}{x_1\,x_2\,x_5} = \frac{a_{x_1x_2x_5}}{x_1\,x_2\,(y_1 - x_4)}.$$

At order $1/x_4^2$: $1/x_5 = 1/(y_1 - x_4) = -1/x_4 - y_1/x_4^2 - \cdots$, so the $1/x_4^2$ piece is $-y_1/x_4^2$. Contribution: $-a_{x_1x_2x_5}\,y_1/(x_4^2\,x_1\,x_2)$.

Fingerprint: $y_1/(x_1x_2)$ at order $1/x_4^2$.

Unique source check. Source of y_1 in a tail: only $1/x_5$ at $\mathcal{Z}_4$. (The other non-bare $1/y_2$ has companion x_2, producing x_2 not y_1.) A single-x_5 monomial $a_{x_5X_aX_b}/(x_5X_aX_b)$ at order $1/x_4^2$ has fingerprint $y_1/(X_aX_b)$; provided $X_a, X_b \neq y_1$, it's unique.

$\{x_1, x_2\} \subseteq F_4 \setminus \{y_1\} = \{x_1, x_2, x_6, y_3\}$: non-constant fingerprint. Hence

$$\boxed{a_{x_1x_2x_5} = 0.}$$

Cyclic rotation kills the full 6-orbit.

Orbit (d): $\{x_3, x_6, y_1\} = a_{x_3x_6y_1}$ at $\mathcal{Z}_2$ Layer 2

$\mathcal{Z}_2$ substitutions: $x_3 = y_2 - x_2$ (non-bare, comp y_2), $y_3 = x_6 - x_2$ (non-bare, comp x_6), $x_1 = -x_2$ (bare). Special: x_2. Free: $F_2 = \{x_4, x_5, x_6, y_1, y_2\}$.

The target on $\mathcal{Z}_2$:

$$\frac{a_{x_3x_6y_1}}{x_3\,x_6\,y_1} = \frac{a_{x_3x_6y_1}}{(y_2 - x_2)\,x_6\,y_1}.$$

At order $1/x_2^2$: $1/x_3$ contributes $-y_2/x_2^2$, so the coefficient is $-a_{x_3x_6y_1}\,y_2/(x_6\,y_1)$.

Fingerprint: $y_2/(x_6\,y_1)$ at order $1/x_2^2$.

Unique source check. Source of y_2: only $1/x_3$'s tail. A single-x_3 monomial $a_{x_3X_aX_b}/(x_3X_aX_b)$ at order $1/x_2^2$ has fingerprint $y_2/(X_aX_b)$, unique provided $X_a, X_b \neq y_2$.

Special check: could X_a or $X_b = y_3$ contribute? $1/y_3 = -1/x_2 - x_6/x_2^2 - \cdots$ has x_6 (not $1/x_6$) in its Laurent tail; so y_3-as-factor puts x_6 in the numerator, not $1/x_6$ in the denominator. The fingerprint $y_2/(x_6\,y_1)$ requires $1/x_6$ directly, so this requires x_6 as a direct factor (not a substituted chord). Since $x_6 \in F_2$, this is possible only with $X_a = x_6, X_b = y_1$ (or vice versa) — giving the target monomial.

$\{x_6, y_1\} \subseteq F_2 \setminus \{y_2\} = \{x_4, x_5, x_6, y_1\}$: non-constant fingerprint.

$$\boxed{a_{x_3x_6y_1} = 0.}$$

Cyclic rotation: orbit has size 3 (the σ^3 stabilizer fixes each element since σ^3 sends $\{x_3, x_6\} \mapsto \{x_6, x_3\} = \{x_3, x_6\}$).

Orbit (e): $\{y_1, y_2, y_3\} = a_{y_1y_2y_3}$ at $\mathcal{Z}_1$ Layer 2

At $\mathcal{Z}_1$: non-bare $y_2 = x_5 - x_1$ with companion x_5. Special: x_1. Free: $F_1 = \{x_3, x_4, x_5, y_1, y_3\}$.

The target on $\mathcal{Z}_1$:

$$\frac{a_{y_1y_2y_3}}{y_1\,y_2\,y_3} = \frac{a_{y_1y_2y_3}}{y_1\,(x_5 - x_1)\,y_3}.$$

At order $1/x_1^2$: $1/y_2$ contributes $-x_5/x_1^2$; the coefficient is $-a_{y_1y_2y_3}\,x_5/(y_1y_3)$.

Fingerprint: $x_5/(y_1y_3)$ at order $1/x_1^2$.

Unique source check. Source of x_5 in a Laurent numerator: only $1/y_2$'s tail. Single-y_2 monomial $a_{y_2X_aX_b}/(y_2X_aX_b)$ at order $1/x_1^2$ gives fingerprint $x_5/(X_aX_b)$; unique provided $X_a, X_b \neq x_5$. $\{y_1, y_3\} \subseteq F_1 \setminus \{x_5\} = \{x_3, x_4, y_1, y_3\}$.

$$\boxed{a_{y_1y_2y_3} = 0.}$$

Orbit has size 1: fixed by σ because the y-set is shift-invariant.

5.8 Completing the proof at $n = 6$

Theorem 3 (Locality at six points)**.** *The six hidden one-zeros $\mathcal{Z}_1, \ldots, \mathcal{Z}_6$ force every non-local coefficient of the general rational ansatz at $n = 6$ to zero. The surviving 14 coefficients are exactly the tree triangulations of the hexagon.*

Proof. Let B be the general rational ansatz, vanishing on each $\mathcal{Z}_r$.

Step 1: Layer 1 covers 129 non-local coefficients. At each $\mathcal{Z}_r$, the $x_r \to \infty$ limit forces every multiset $M \subseteq F_r$ to have $a_M = 0$ (Lemma 3). By the counting in Section 5.4, the union over r of $\{M : M \subseteq F_r\}$ contains exactly 129 distinct non-tree multisets.

Step 2: Blocking-set lemma. By Lemma 5, no tree is contained in any F_r, so Step 1 does not accidentally kill any tree.

Step 3: Layer 2 covers the remaining 22. The 22 multisets failing Layer 1 partition into 5 cyclic orbits (Proposition 1). Sections 5.7 verifies that one representative of each orbit is killed by Layer-2 companion-isolation at a specific zone. By cyclic symmetry, every member of each orbit is killed: applying σ^k to both the multiset and the zone transforms a verified kill at $\mathcal{Z}_r$ into a kill of the σ^k-image multiset at $\mathcal{Z}_{r+k}$.

Step 4: Conclusion. All $129 + 22 = 151$ non-local coefficients of B are forced to zero. The surviving 14 are the tree triangulations, satisfying the tree-amplitude relations coming from $A_6|_{\mathcal{Z}_r} = 0$. □

Remark on Layer 3. A "Layer 3" analysis at order $1/x_r^4$ (double companion-isolation, using monomials with *both* non-bare substitutions as factors) does produce additional kills at each zone (3 per zone at $n = 6$). However, these kills are redundant — every Layer-3-killable multiset is already in some $F_{r'}$ and hence killed by Layer 1 at $\mathcal{Z}_{r'}$. At $n = 6$, Layers 1 and 2 alone suffice. Layer 3 may become essential at higher n.

6 Looking ahead

The proof template generalizes. At each zone $\mathcal{Z}_r$ at general n:

- $|F_r| = \binom{n}{2} - n - (n-3) - 1 = \frac{n(n-3)}{2} - (n-2)$.
- The number of non-bare substitutions is $n - 4$, each with a companion in F_r.
- Layer 1 kills $\binom{|F_r|+n-4}{n-3}$ multisets at order x_r^0.
- Layer k ($k \geq 1$) kills multisets with exactly k non-bare indices via k-fold companion-isolation at order $1/x_r^{2k}$.

Layer counts per zone:

n	$\lvert F_r \rvert$	L0	L2	L4	L6	Total per zone
4	0	0	—	—	—	0
5	2	3	1	—	—	4
6	5	35	20	3	—	58
7	9	495	360	84	6	945

At $n = 6$ our explicit union-count gives 129 from Layer 1 and 22 from Layer 2, a union of 151. The detailed orbit structure (Section 5.6) explains *why* these are exactly the 151 non-local multisets.

A general-n proof along these lines would require:

1. A generalization of the blocking-set lemma: at every zone and every n, no triangulation lies in F_r.
2. An orbit enumeration of Layer-1 failures (relating to which multisets escape every F_r).
3. Explicit companion-isolation arguments for each orbit representative, using the appropriate Layer.

The class hierarchy of chords (short, long, extra-long, ...) — with each new class born at $n = 2k$ — refines the bookkeeping. We leave the general-n proof for future work.